Blinkit Sales Analysis: Project Report

Author: Shaik Fazila Zainab | **Tools:** Power BI, Excel | **Repository:** Blinkit_Analysis

1. Objective

Blinkit sells groceries through a network of outlets that differ in type, size, and location. This project answers one business question: **which outlet types, locations, and product categories drive sales, and where should the business focus?** The result is an interactive Power BI dashboard plus the findings in this report.

2. Dataset

The source is *BlinkIT Grocery Data* (Excel), with 8,523 records and 12 columns:

- **Item details:** Item Identifier, Item Type, Item Fat Content, Item Weight, Item Visibility
- **Outlet details:** Outlet Identifier, Outlet Establishment Year, Outlet Size, Outlet Location Type, Outlet Type
- **Performance:** Sales, Rating

3. Data Preparation

- **Item Fat Content** had inconsistent labels (for example "LF" and "low fat" next to "Low Fat" and "Regular"). These were standardized into two categories: Low Fat and Regular.
- **Item Weight** has missing values. It was not used in the dashboard metrics, so no values were imputed.
- The cleaned data was loaded into Power BI, where the KPIs, visuals, and slicers were built.

4. Key Performance Indicators

KPI	Value
Total Sales	$1.20M
Average Sales per Item	$141

KPI	Value
Average Rating	3.9
Number of Items	8,523

5. Analysis

5.1 Performance by Outlet Type

Outlet Type	Total Sales	Items	Avg Sales	Avg Item Visibility
Supermarket Type1	$787.55K	5,577	$141	0.06
Grocery Store	$151.94K	1,083	$140	0.10
Supermarket Type2	$131.48K	928	$142	0.06
Supermarket Type3	$130.71K	935	$140	0.06

Supermarket Type1 generates 65.5% of sales and also holds 65.4% of the items. Each outlet type's share of sales almost exactly matches its share of items, and average sales per item sit between $140 and $142 for all four types.

5.2 Performance by Outlet Location

Location Tier	Total Sales	Share of Sales
Tier 3	$472.13K	39.3%
Tier 2	$393.15K	32.7%
Tier 1	$336.40K	28.0%

5.3 Performance by Outlet Size

Medium outlets led with about $508K, followed by Small outlets at $444.79K and High-size outlets at $248.99K.

5.4 Performance by Item Fat Content

Low Fat items contributed $776.32K (64.6% of sales) and Regular items $425.36K (35.4%). Low Fat items lead in all three location tiers.

5.5 Performance by Item Type

Fruits and Vegetables and Snack Foods are the top categories at about $0.18M each, together close to 30% of sales. Household ($0.14M), Frozen Foods ($0.12M), and Dairy ($0.10M) follow, so the top five categories make up about 60% of sales. Seafood, Breakfast, and Starchy Foods sit at the bottom at $0.01M to $0.02M.

5.6 Outlet Establishment Year

Outlets established in 2018 produced the highest sales at $205K. Most other establishment years fall in a narrow band of $129K to $133K, and 2011 is the lowest at $78K.

6. Key Insights

1. **Sales are driven by volume, not price.** Average sales per item are almost identical across outlet types (about $140), so Supermarket Type1 leads because it stocks and sells far more items.
2. **Tier 3 locations lead sales** at 39.3%, ahead of Tier 2 and Tier 1.
3. **Two categories carry the business.** Fruits and Vegetables plus Snack Foods account for about 30% of sales.
4. **Low Fat items outsell Regular items** by about 1.8 to 1 ($776K vs $425K).
5. **Item visibility does not appear to lift sales.** Grocery Stores have the highest average visibility (0.10) yet the same average sales per item as other outlet types.
6. **2018 is an outlier.** Outlets established that year sell about 55% more than the typical year.

7. Recommendations

These are suggestions based on the patterns above and should be tested before acting on them.

- Prioritize availability and promotions in Fruits and Vegetables and Snack Foods, the two biggest sales drivers.

- Treat Tier 3 locations and the Supermarket Type1 format as the strongest growth bets, since they lead in sales.
- Check why 2018-established outlets perform so much better (location mix, size, or assortment) and see whether it can be repeated.
- Review stock levels of Low Fat versus Regular items by outlet to avoid shortages in the better-selling group.
- Test shelf-visibility changes in a small pilot before investing, because visibility shows no clear link to sales in this data.

8. Limitations

- The data has no date column, so there is no time-trend or seasonality analysis. Only the outlet establishment year is available.
- Records are item-and-outlet level, not individual customer transactions.
- The analysis is descriptive. The patterns show association, not cause.
- Average rating varies little across segments, so it adds limited insight.
- Item Weight has missing values and was left out of the analysis.

9. Conclusion

Blinkit's sales are concentrated in a few places: Supermarket Type1 outlets, Tier 3 locations, and two item categories. Outlet types earn similar revenue per item, so growth comes from selling more items in the right places and categories. Next steps could include adding SQL-based validation of the KPIs and a time dimension if transaction dates become available.